Practical Assignment

Experimental Comparison of PSO and Genetic Algorithms in Real-Value Representations

1. Objective

The goal of this assignment is to implement and experimentally compare Particle Swarm Optimization (PSO) and Real-coded Genetic Algorithm (GA).

We consider the continuous optimization problem:

$$\min_{\mathbf{x} \in [lb, ub]^d} f(\mathbf{x})$$

where:

- d is the dimensionality of the problem
- lb and ub define lower and upper bounds
- $\mathbf{x} \in \mathbb{R}^d$

The functions considered in this assignment are multimodal and contain multiple local optima.

1.1 Ackley function

Domain:

$$x_i \in [-32.768, 32.768]$$

Global optimum:

$$f(\mathbf{x}^*) = 0 \quad \text{at} \quad \mathbf{x}^* = (0, \ldots, 0)$$

Expression:

$$f(\mathbf{x}) = -20 \exp\left(-0.2\sqrt{\frac{1}{d}\sum x_i^2}\right) - \exp\left(\frac{1}{d}\sum \cos(2\pi x_i)\right) + 20 + e$$

Characteristics:

- multimodal landscape
- flat outer region
- narrow basin near optimum
- useful to evaluate exploration vs exploitation

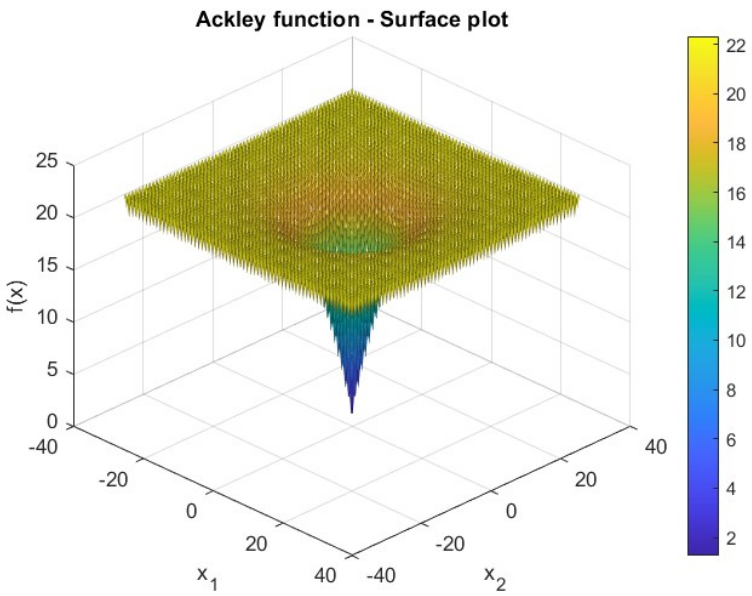

Figure 1: Rastringin function

1.2 Rastrigin function

Domain:

$$x_i \in [-5.12, 5.12]$$

Global optimum:

$$f(\mathbf{x}^*) = 0 \quad \text{at} \quad \mathbf{x}^* = (0, \ldots, 0)$$

Expression:

$$f(\mathbf{x}) = 10d + \sum_{i=1}^{d} \left(x_i^2 - 10\cos(2\pi x_i)\right)$$

Characteristics:

- highly multimodal

- large number of regularly distributed local minima
- challenging landscape for population-based methods

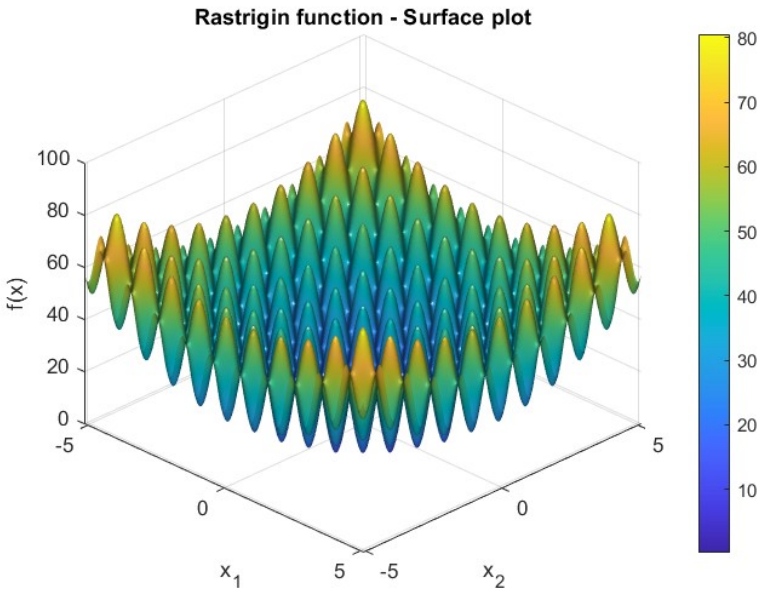

Figure 2: Rastringin function

2. Tasks

2.1 Function visualization

Visualize both functions in two dimensions ($d = 2$):

- 3D surface plot
- contour plot

Discuss how the shape of the landscape may influence search difficulty.

2.2 Implementation of PSO

Implement Particle Swarm Optimization using real-valued representation.

Each particle must include:

- position vector
- velocity vector

- personal best position (pbest)
- global best position (gbest)

Velocity update:

$$\mathbf{v}_i^{t+1} = w\mathbf{v}_i^t + c_1\mathbf{r}_1 \odot (\mathbf{pbest}_i - \mathbf{x}_i^t) + c_2\mathbf{r}_2 \odot (\mathbf{gbest} - \mathbf{x}_i^t)$$

Position update:

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}$$

Include:

- bound handling
- optional velocity clamping
- convergence tracking

2.3 Implementation of real-coded GA

Implement a genetic algorithm using floating-point representation. A personal proposal of:

- real-valued crossover
- real-valued mutation

Note: Binary encoding is not allowed.

2.4 Experimental comparison

Compare both algorithms on:

- Ackley (optional)
- Rastrigin

Dimensions:

$$d \in \{10, 30\}$$

2.5 Experimental protocol

To ensure a fair comparison:

- same population size
- same search domain
- multiple independent runs (at least 100 times)

Optional success criterion:

$$f(\mathbf{x}) < 10^{-3}$$

2.6 Analysis

Discuss:

1. convergence speed differences
2. robustness to random initialization
3. effect of dimensionality
4. premature convergence behaviour
5. exploration vs exploitation trade-off
6. sensitivity to multimodality

Interpret results rather than only presenting plots.

3. Implementation guidelines

Efficient MATLAB implementation is expected (Appropriate Stopping Conditions for the main loop are mandatory).

Further recommendations:

- vectorize computations
- avoid unnecessary loops
- evaluate populations in batch
- separate operators and evaluation

4. Deliverables

Submit in a zip file:

1. MATLAB source code
2. readme file to run the code
3. short report in pdf (max 2 pages)